LAXMI NARAYANA YAGA

laxminarayana.y@mycvscout.com • (314) 305-8989 • LinkedIn • St. Louis, MO (Open to relocation)

PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience designing, training, and deploying machine learning and generative AI solutions across cloud environments. Skilled in building end-to-end pipelines using Python, PyTorch, TensorFlow, and AWS SageMaker, with expertise in LLMs, RAG, LangGraph, and LangChain for intelligent automation and knowledge retrieval. Experienced in feature engineering, model optimization, and MLOps for scalable deployment. Passionate about developing explainable, secure, and production-grade AI systems that enhance decision-making, automate workflows, and deliver measurable business impact.

SKILLS

Programming Languages & Tools: Python, R, SQL, Git, Jupyter Notebook, Google Colab, Docker, Spark, PySpark
Machine Learning: Supervised & Unsupervised Learning, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, K-Means, DBSCAN, Principal Component Analysis (PCA), Feature Engineering, Ensemble Methods, Time Series Forecasting, XGBoost, AutoML workflows, LLMOps (large language model operations)
Deep Learning: Neural Networks, ANN, CNN, RNN, Generative Models (GANs, VAEs), Natural Language Processing (NLP), Multimodal AI
Deep Learning Frameworks: TensorFlow, Keras, PyTorch, Hugging Face Transformers
AI & Generative AI Technologies: LangChain, LangGraph, LLaMA, Mistral, LlamaIndex, Retrieval-Augmented Generation (RAG), Agentic AI
NLP & Large Language Models: BERT, ROBERTa, GPT-4, Claude, Named Entity Recognition (NER), NLTK, Spacy, Prompt Engineering
Data Visualization & Libraries: Plotly, Tableau, Power BI, Scikit-learn, Pandas, NumPy, SciPy, Dashboarding with AI-augmented analytics
Databases: SQL Server, MySQL, PostgreSQL, MongoDB
Cloud & MLOps: AWS (SageMaker, Lambda, CloudWatch, CloudFormation), GCP, Azure, Model Deployment, Flask, FastAPI, Streamlit, CI/CD

WORK EXPERIENCE

PNC Financial Services — **Missouri, USA**
AI/ML Engineer — September 2024 - Present

- Developed and deployed credit risk prediction models using Python, PySpark, and AWS SageMaker, automating real-time scoring of loan applications and improving underwriting accuracy by 28% through AutoML workflows and optimized feature pipelines.
- Implemented a Retrieval-Augmented Generation (RAG) framework with LangGraph, LlamaIndex, and LLaMA, enabling financial advisors to query internal documents and generate regulatory-compliant insights, cutting manual data retrieval time by 60%.
- Built and maintained end-to-end MLOps pipelines integrating TensorFlow models with AWS Lambda and SageMaker endpoints, ensuring reproducibility, continuous deployment, and 99.5% model uptime across production environments.
- Trained and fine-tuned multimodal and Generative AI models (GANs, VAEs) for fraud detection and document classification, leveraging Agentic AI frameworks and Claude for automated anomaly explanation and case summarization, reducing manual review effort by 35%.
- Engineered data pipelines using PySpark and PostgreSQL to cleanse, aggregate, and align transactional and behavioral data, enabling high-fidelity datasets for model training and supporting data governance compliance under FFIEC standards.
- Designed and visualized AI-augmented analytics dashboards in Tableau, combining model outputs, key metrics, and explainability layers to enhance executive decision-making and transparency in lending and risk management operations.

Zensar Technologies — **India**
ML Engineer — August 2021 – July 2023

- Developed and deployed supervised and unsupervised learning models using Python, Scikit-learn, and XGBoost, improving customer churn prediction accuracy by 24% through advanced feature engineering and model optimization.
- Implemented clustering algorithms such as K-Means and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to segment customers based on behavioral data, enabling personalized marketing strategies and reducing campaign costs by 18%.
- Applied Principal Component Analysis (PCA) to reduce dataset dimensionality by 40%, improving model interpretability and accelerating training performance for production workflows.
- Built and fine-tuned Convolutional Neural Networks (CNNs) and Bidirectional Encoder Representations from Transformers (BERT) models using Keras and PyTorch, automating image recognition and text classification tasks with 92% accuracy.
- Developed custom Natural Language Processing (NLP) pipelines using Natural Language Toolkit (NLTK) and spaCy for entity extraction, sentiment analysis, and document summarization, increasing text-processing throughput by 3×.
- Designed and automated end-to-end model deployment pipelines using FastAPI and Azure Machine Learning, cutting deployment time by 50% and enabling scalable API-based inference services.
- Integrated ML-driven insights into Power BI and Plotly dashboards, enabling real-time visualization of key performance indicators and improving business decision-making accuracy.
- Optimized data pipelines using Pandas, NumPy, and MongoDB to streamline data preprocessing and storage for large-scale analytics

EDUCATION

Master of Science in Computer and Information Sciences | Saint Louis University, St. Louis, MO — **May 2025**
Bachelor of Technology in Information Technology | Institute of Aeronautical Engineering, Hyderabad, India — **May 2023**

CERTIFICATIONS

Develop Generative AI solutions with Azure OpenAI Service | Microsoft Azure AI Fundamentals: Generative AI | Microsoft Azure Fundamentals: Describe cloud concepts | Anaplan Level 1&2 Model Builder | Become a Data Scientist: SQL, Tableau, ML & DL [4-in-1] | Hands on Machine Learning and Deep Learning